

1. Solution. By the Central Limit Theory for large samples, we know that the sampling distribution of the sample mean is approximately normally distributed since the sample size $n=64 \geq 30$. The standard deviation of the population $\sigma=8$ is known, thus ~~we~~ Z-distribution is used for confidence interval estimation of the population mean μ ,

$$\begin{aligned} [\bar{x} \pm Z_{0.025} \sigma_{\bar{x}}] &= [\bar{x} \pm Z_{0.025} \frac{\sigma}{\sqrt{n}}] \\ &= [85 \pm 1.96 \cdot \frac{8}{\sqrt{64}}] \\ &= [83.04, 86.96]. \quad \# \end{aligned}$$

2. Solution. The probability that the true average monthly sales of a product is contained in $[170,000, 200,000]$ is 95%. $\#$.

3. Solution. The standard deviation of the population is unknown, thus t-distribution is used for confidence interval estimation for the population mean,

$$\begin{aligned} \text{Set 1: } [\bar{x}_1 \pm t_{0.025} \frac{S_1}{\sqrt{n}}] &\stackrel{df=7}{=} [\frac{36}{8} \pm 2.365 \frac{\sqrt{14}}{\sqrt{8}}] \\ &= [1.3714, 7.6286] \end{aligned}$$

$$\begin{aligned} \text{Set 2: } [\bar{x}_2 \pm t_{0.025} \frac{S_2}{\sqrt{n}}] &\stackrel{df=7}{=} [\frac{36}{8} \pm 2.365 \frac{\sqrt{6}}{\sqrt{8}}] \\ &= [2.4518, 6.5482] \end{aligned}$$

Although they have the same mean $\bar{x}$ and range, the sample standard deviations are different, and thus results in different confidence intervals. $\#$.

4. Solution: (a) Since the sample size $n=50 \geq 30$ and the standard deviation of the population is known, a 99% confidence interval estimate of the true population average amount of paint included in a 1-gallon can is

$$\begin{aligned} [\bar{x} \pm Z_{0.005} \sigma_{\bar{x}}] &= [\bar{x} \pm 2.575 \frac{\sigma}{\sqrt{n}}] \\ &= [0.995 \pm 2.575 \frac{0.02}{\sqrt{50}}] \\ &= [0.9877, 1.0023]. \end{aligned}$$

(b) According to the CI obtained above, we can conclude that there is a 0.99 probability that the true average amount of paint is contained in the interval $[0.9877, 1.0023]$. Therefore, I do not think that the store owner has a right to complain.

(c) No. Since the sample size $n=50 \geq 30$, and by the central limit theory for large samples, the sampling distribution of the sample mean is approximately normally distributed.

(d) If we use a 95% confidence interval estimation

$$\begin{aligned} [\bar{x} \pm Z_{0.025} \sigma_{\bar{x}}] &= [\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}] \\ &= [0.995 \pm 1.96 \cdot \frac{0.02}{\sqrt{50}}] \\ &= [0.9895, 1.0005], \end{aligned}$$

I still do not think that the store owner has a right to complain. #.

5. Solution (a) The standard deviation of the population is unknown, thus t -distribution is used for confidence interval estimation of the population average tread wear index,

$$\left[\bar{x} \pm t_{0.025} \frac{S}{\sqrt{n}} \right] \xrightarrow{df=17} \left[195.3 \pm 2.11 \cdot \frac{21.4}{\sqrt{18}} \right]$$

$$= [184.6571, 205.9429]$$

(b) I do not think that the consumer organization should accuse the manufacturer of producing tires that do not meet the performance information provided on the sidewall of the tire. According to the CI above, we can be 95% confident that the population average tread wear index is contained in the interval $[184.6571, 205.9429]$ which contains 200.

(c) Since the confidence interval above is an estimation for the population average, it is reasonable that an observed tread wear index of 210 for a particular tire is outside the interval. ✗

6. Solution (a) The standard deviation of all greeting cards is unknown, thus t -distribution is used for a 95% confidence interval estimate of the average value,

$$\left[\bar{x} \pm t_{0.025} \frac{S}{\sqrt{n}} \right] \xrightarrow{df=19} \left[1.67 \pm 2.093 \frac{0.32}{\sqrt{20}} \right]$$

$$= [1.5202, 1.8198]$$

(b). According to the above result, we have 95% confidence that the true average value of all greeting cards locates between 1.5202 and 1.8198. Thus, we have 95% confidence that the total value is between $1.5202 \times 20 = 30.4$ and $1.8198 \times 20 = 36.4$ ✗.

7. Solution. The sample proportion $\hat{p} = \frac{30}{100} = 0.3$. Since both $n\hat{p}$ and $n(1-\hat{p})$ are larger than 5, the proportion is approximately normally distributed $\therefore$ Hence a 90% confidence interval estimate of the population proportion is

$$\left[\hat{p} \pm Z_{0.05} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] = \left[0.3 \pm 1.645 \sqrt{\frac{0.3 \cdot 0.7}{100}} \right] \\ = [0.2246, 0.3754] \#$$

8. Solution (a) The sample proportion $\hat{p} = \frac{80}{200} = 0.4$. Since both $n\hat{p}$ and $n(1-\hat{p})$ are larger than 5, the population proportion is approximately normally distributed. Hence a 99% confidence interval estimate of the population proportion is

$$\left[\hat{p} \pm Z_{0.005} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] = \left[0.4 \pm 2.575 \sqrt{\frac{0.4 \cdot 0.6}{200}} \right] \\ = [0.3108, 0.4892]$$

(b) According to the result above, ~~less~~ ^{we have 99% confidence that} than 50% of claims are paid within 2 months of their receipt. The proportion is much lower than 90% which is required. #

9. Solution. Since the standard deviation is known, Z-distribution is used for determining the sample size,

$$n = \left(\frac{Z_{0.025} \sigma}{E} \right)^2 = \left(\frac{1.96 \times 400}{50} \right)^2 = 245.86 \\ \approx 246.$$

Therefore, sample size $n = 246$ is necessary. #

10. Solution. (a) In order to achieve a sample precision of $\pm 2\%$ with 99% confidence, the sample size should be

$$\begin{aligned} n &= p(1-p) \left(\frac{Z_{0.005}}{0.02} \right)^2 \\ &= 0.05(1-0.05) \cdot \left(\frac{2.575}{0.02} \right)^2 \\ &= 787.4 \approx 788 \end{aligned}$$

(b) If the maximum tolerable error rate is 10%, the sample size should be

$$\begin{aligned} n &= p(1-p) \left(\frac{Z_{0.005}}{0.02} \right)^2 \\ &= 0.1 \times (1-0.1) \left(\frac{2.575}{0.02} \right)^2 \\ &= 1491.9 \approx 1492 \end{aligned}$$

(c) If the precision is $\pm 5\%$, the sample size should be

$$\begin{aligned} n &= p(1-p) \left(\frac{Z_{0.005}}{0.05} \right)^2 \\ &= 0.05(1-0.05) \left(\frac{2.575}{0.05} \right)^2 \\ &= 125.98 \approx 126. \end{aligned}$$